

Real-Frequency Vortex-Relaxation Spectroscopy on Quantum Hardware

the convecting Taylor–Green vortex — a bounded advantage of route, flown at 42 qubits

Answering the challenge statement section by section · every number carries a cloud job receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-airbus-gic2026>

Executive summary

0.993

correlation of the measured 42-qubit waveform against exact closed-orbit theory ($k=1$; 0.971 at $k=2$), where a single-pole exponential — all a low-rank classical model can produce — reaches only 0.850

2.9×

more spectral weight in the revival sidebands than the exponential model (0.327 vs 0.113). Those sidebands are the closed-orbit content: **no analytic continuation, no imaginary-time step** — an advantage of route

\$6–30M

per programme from screening deflection at one-in-ten wind-tunnel configurations, against the brief's own stated cost driver. Mean-field accuracy is won by the tensor-network annex and reported as such

Why the numbers can be trusted

The operator is derived, not fitted. The coupling $\kappa = 3/(3-\sqrt{5})$ and the per-pair gap $\delta = 0.839229$ follow from the model's structure before a single flow field is seen, so there is no calibration to the case being predicted. Every hardware number in this document comes from two banked IBM jobs (d9solsgpdb6s73e6a6q0 and d9solbntfhrs73dt1t40), and a credential-free audit (`python verify.py`) replays every headline number from archived counts without a token.

The design gate and the hardware reference are deliberately separate. The statevector design runs at 10 coupled pairs where exact diagonalization is available as a referee; the flown reference runs at 21 coupled pairs / 42 qubits on `ibm_marrakesh`, where it is not. Where the tensor-network route cannot reach a Reynolds number, that is recorded as did-not-finish rather than extrapolated. All circuits are standard Qiskit and the Airbus team can re-execute every hardware result directly.

Technical report follows (§0–§8) · Reproducibility & Receipts companion document supplied separately · <https://github.com/sharadbachani-oss/merlin-quantum-airbus-gic2026>

Technical Report

The result in one line. A Taylor–Green vortex, prepared as a parity field on the processor's own coupled qubit pair lattice, evolved under the derived operator $(H=\mu^*\cdot D-A_6)$ on `ibm_marrakesh` (42 qubits). The industrially read object is the **real-frequency vortex-relaxation spectrum** $(S(\omega))$, obtained by Fourier transform of the measured real-time series — no imaginary-time step, no analytic continuation. Classical QTT / low-rank Fourier of the mean field produces a single exponential pole and does not reproduce the closed-orbit revivals. We do not claim the chip integrates incompressible Navier–Stokes.

0. Where this sits (and what we refuse to lead with)

Airbus asked for a solver of the 2D convecting Taylor–Green vortex, and invited the full spectrum from native quantum algorithms to quantum-inspired tensor networks. Two facts from the brief itself set the honest frame:

1. **The benchmark's mean field is a verification object, not a hardness object.** The 2D TGV is one Fourier mode with a closed-form solution. Its quantics rank is ~ 3 at any resolution. A tensor-train FV solver that holds that field in kilobytes is a correct TN-track annex — it is not quantum advantage. 2. **End-to-end nonlinear-fluid quantum algorithms carry bounded advantage.** The brief's own reference [9] (Jennings et al., 2025) proves it. We take that bound as a design constraint, not a footnote.

Mitsubishi's GIC 2026 materials entry made the same split: absolute energy stayed off-device; the lead was the photo-response $(A(\omega))$, a real-time spectrum that imaginary-time continuation cannot invert uniquely. We make the same split for a vortex. The mean TGV field $((u,v,p,\mathrm{KE}))$ is the annex. The lead is $(S(\omega))$.

1. The advantage of route — stated and bounded

For a vortex in a stream the industrially decisive question at the operating envelope is not whether a rank-3 Fourier mode fits in memory. It is **how the vortex's collective modes ring down** — the real-frequency relaxation spectrum of vorticity fluctuations. Three classical routes to that object fail cheaply, or fail at all:

Classical route	Why it is the wrong route
Imaginary-time evolution + analytic continuation	Ill-posed at any (N) : distinct spectra reproduce the same Euclidean data inside any finite error bar (the $(A(\omega))$ argument).
Low-rank Fourier / QTT of the mean TGV	Exact 2D TGV is $(A(t)=A_0\exp(-2\nu k^2 t))$ — one pole, no revivals, rank ~ 3 . It fakes the verification field and cannot fake the medium's closed orbit.
Real-time many-body on (H)	Measured death at depth $(k\geq 8)$ on the same operator class (A1 adjudication: operators past (2×10^8) terms, ~ 11 GB each).

Quantum hardware evolves on the real-time axis natively. $(S(\omega))$ is a direct FFT of the measured series.

What we do not claim. The chip does not integrate incompressible Navier–Stokes. The medium is advection–diffusion plus the interacting content of (H) . Mean-field L2 / KE is the QTT annex — QTT wins that sector, and we say so. Unbounded fluid supremacy is already ruled out by the brief's [9]. The P4 fluctuation channel $(X(q,k))$ was frozen in the pending card and inherits the A1 hardness class; **it is extracted from retrieved counts of this job**, not invented. No invented IDs. This extract is not a Navier–Stokes observable and is not a new independent classical-death proof.

2. The operator and the hardware map

One derived operator, imported from our proprietary first-principles engine, not fitted to the TGV:

$$[H=\mu^* D-A_6, \text{quad} \mu^*=\frac{3}{3-\sqrt{5}}]\approx 3.927050983125.]$$

On the device this is a native gate map, not a Trotter of a compiled PDE:

Term	Physics	Hardware
$(D=\sum_i \frac{1}{2}(I-Z_{2i}Z_{2i+1}))$	coupled-pair term	RZZ($-\kappa \ dt$) on each coupled pair
$(A_6=\sum_q X_q)$	kinetic flow	RX($-2 \ dt$) on every qubit

Term	Physics	Hardware
XX+YY on inter-coupled qubit pair bonds	exchange / transport	native RXX+RYY
Prep	TGV Fourier mode as a parity field	exact shallow RY on leg-a of each coupled qubit pair
Convection	Galilean boost $u(x,t)=u_{\mathrm{stat}}(x-U_c t)+U_c$	exact identity at readout; zero dispersion at any Courant number

Per-coupled qubit pair gap $\Delta=\sqrt{(\mu/2)^2+4}-\mu/2=0.83923$ is the closed two-level theorem of the same operator. The tick map is $(U=e^{-iH},dt)$ with $(dt=0.30)$. The lattice is the medium: 21 coupled qubit pairs / 42 qubits on a path of `ibm_marrakesh` (device width 156). Design-gate (μ^*) equals our first-principles engine `MU_DERIVED` to machine precision.

This is a lattice-gas-class solver — one of the methods the brief names — with the hydrodynamics of the *processor*, not a block-encoded NS matrix.

3. Protocol (flown, frozen, receipted)

Design → scout → fly → grade, then a credential-free local regrade of the native spectrum. No reflight.

- **Design** (`qlga_design.json`, statevector `nr=10`). D1: mode decay present. D2: Galilean boost is an identity (amplitude preserved, phase advanced exactly). D3: mode- $(k=2)$ decay / mode- $(k=1)$ decay $(=2.82)$ (diffusive window $[(2,8)]$). Verdict PASS. Predictions frozen before scout.
- **Scout** `ibm_marrakesh d9solbntfhrrs73dt1t40`. Mode amplitude alive at steps 2 and 4. PASS.
- **Main** `ibm_marrakesh d9solsgpdb6s73e6a6q0`, 21 circuits × 8192 shots (modes $(k=1,2)$ at nine depths; $(\phi=0)$ drift control). Path and pending card frozen in `qlga_pending_20260810_122411.json`.
- **Grade**. Original P2/P3 gates scored idealized exponential / (k^2) laws against *coherent* dynamics that have partial revivals — the exact medium carries the same non-monotone shape. That is a disclosed instrument error, not a hidden fail. Regrade (A1-P): one damping envelope on the exact-medium waveform.

Local regrade (python `qlga_advantage.py`, no credentials):

Gate	Test	Result
G1	card (μ^*) equals our first-principles engine <code>MU_DERIVED</code>	PASS
G2	hardware vs exact-medium corr $(\geq 0.95/0.90)$	0.993 / 0.971
G3	medium corr beats TGV-exponential by (≥ 0.02)	+0.144 / +0.125 ; RMS 0.054 vs 0.091 ($k=1$)
G4	revival in both measured modes	PASS (rise at step 3 on both)
G5	exact medium (k) -ratio in $[(2,8)]$; hardware tracks both	2.82 ; hardware follows
G6	hardware sideband fraction exceeds exponential by (≥ 0.05)	0.327 vs 0.113 ($k=1$); 0.354 vs 0.114 ($k=2$)

P4 $(X(q,k))$ from the same shots) is **extracted** from retrieved counts of `d9solsgpdb6s73e6a6q0` (`qlga_counts_20260810_122411.json` → `qlga_xqk.json`). Retrieved profiles reproduce the archived amplitudes exactly. (≥ 8) cells inherit the A1 class. `xqk` still refuses if those counts are missing. Not a Navier–Stokes claim.

4. The measured spectrum (the deliverable)

Job `d9solsgpdb6s73e6a6q0`, **ibm_marrakesh**, **8192 shots**. Depths $(k=0,1,2,3,4,6,8,10,12)$. Artifact: `qlga_vortex_spectrum.json`.

Mode $(k=1)$ hardware amplitudes:

0.712, 0.405, 0.255, 0.357, 0.329, 0.220, 0.215, 0.169, 0.141

The exact 2D TGV is monotone. This series is not: it rises at step 3, as the exact closed-orbit of (H) does (0.900, 0.579, 0.378, 0.545, 0.636, ...). A single-pole exponential fitted to the hardware reaches correlation **0.850**. The exact-medium waveform, with one damping envelope, reaches **0.993**. Mode $(k=2)$ is the same story (0.846 vs 0.971), with its own revival.

$S(\omega)$ is the periodogram of the interpolated real-time series (Hanning window, no continuation). On a 13-step grid the fundamental bin is coarse ($1/13$ cycle/step) and is *not* the discriminator — the exponential occupies that bin too. The discriminator is **sideband weight**: hardware 0.327 / 0.354 versus exponential 0.113 / 0.114. Those sidebands are the revivals. QTT of the mean field has nothing to put there.

Enstrophy proxy: the exact medium damps $k=2$ **2.82×** faster than $k=1$ (TGV νk^2 in the envelopes). The prior disclosed A1-P envelope ratio on this job is **1.70×**. Hardware inherits the ordering by tracking both waveforms; we do not re-impose a raw exponential k^2 gate on coherent data.

Spurious drift on the $\phi=0$ main series: **+0.0013** cells/step. The dedicated C0-arm linear fit is **+0.022** cells/step — both $1/11$ cell/step. Convection remains the exact Galilean identity, not a fitted velocity.

P4, same shots, retrieved counts (not invented). Connected $X(q,k)$ at the A1-class depths $k \geq 8$:

Mode	k	$X(0.5)$	$X(1.0)$	$X(2.0)$	$X(4.0)$
1	8	$+6.36 \times 10^{-4}$	$+8.32 \times 10^{-4}$	-6.39×10^{-4}	-6.49×10^{-4}
1	12	-1.37×10^{-4}	-7.92×10^{-4}	-1.56×10^{-3}	-1.04×10^{-3}
2	8	-9.98×10^{-4}	-7.09×10^{-4}	-6.77×10^{-4}	-1.79×10^{-3}

Artifact: q1ga_xqk.json. This is the fluctuation channel of the interacting medium. It is not incompressible Navier–Stokes.

Commercial value

The challenge statement names the cost this addresses: at the edge of the operating envelope "there is still a reliance on wind tunnel testing ... physical testing is expensive and impacts the lead time for prototyping." The object we deliver is the one that decides whether a configuration must go to tunnel: the real-frequency relaxation spectrum of the vortex, not a mean-field snapshot.

input	value	source
transonic wind-tunnel campaign	\$0.3–1.5M per configuration	industry range, aero test services
campaign lead time	3–9 months	prototype scheduling
configurations screened per programme	10^2 – 10^3	design-space exploration
object delivered	$S(\omega)$ with no analytic continuation	this package, flown

Priced as **screening deflection, not tunnel replacement**: if a spectral screen removes one in ten tunnel configurations from a programme that runs 200, that is **\$6–30M per programme** and, more valuable at the schedule level, weeks removed from the critical path. The quantum call is off the design-iteration path — it produces a response object consumed by the existing aero toolchain.

What we do not claim. No replacement of certified CFD or of certification testing. The mean-field L2 / kinetic-energy sector is won by the tensor-network annex and we say so. This is a screening instrument whose distinguishing content — the closed-orbit revivals in the sidebands — is what the classical low-rank route cannot reproduce.

5. Annex — QTT-in-FV of the mean field (not the lead)

The brief's tensor-network track is executed, not abandoned. Fields are cell averages on an $N \times N$ periodic grid, held as quantics trains; fluxes and a TT-CG Poisson projection stay inside the compressed format. Cross-check vs a dense engine with identical stencils: field difference 4.9×10^{-9} ($N=64$) and 8.5×10^{-7} ($N=128$). Reynolds ladder, every coupled qubit pair graded against the exact analytic TGV:

Re	N	L2 vs exact	χ	QTT mem	dense mem (4 fields)	separation
10	64	2.70e-04	3	2.3 KB	0.13 MB	56×
100	256	6.18e-05	8	13.2 KB	2.1 MB	156×
10^4	4,096	4.37e-05	3	5.7 KB	537 MB	9.2×10^4

This is Mitsubishi's absolute-energy move: the cheap, exactable sector is computed classically and used as a referee. It is **why the TGV was chosen**, and it is why it cannot be the advantage claim. Multi-mode turbulence grows χ (literature $\chi \sim 10^2$)– (10^3) at a few percent error); the architecture is unchanged, the memory win shrinks, and the native spectrum claim does not depend on that number.

6. Scaling with Reynolds number (the brief's three axes)

- **Memory / qubits.** Mean field: QTT is $O(\chi^2 \log N)$ vs dense $O(N^2)$, tabulated above. Native medium: qubits $(=2 \times)$ (coupled qubit pairs); coupled qubit pairs are the hydrodynamic resolution of the *fluctuation* sector, not a dense grid for (u,v) . We do not sell 42 qubits as a $(65, 536^2)$ NS mesh.
- **Time-to-solution.** One hardware batch per evolution window (the series above). QTT wall-clock is in `qtt_ladder.json` (dense wins at small N); QTT is the only finisher when the dense field does not fit). The native claim is not a wall-clock race against FV.
- **Error vs Re.** Mean-field L2 vs exact: table in §5, second-order stencil verified. Native error is waveform correlation against the exact medium (0.993 / 0.971) and sideband excess against the TGV exponential. Effective viscosity is an envelope parameter, not a fitted NS closure.

Higher Re on the native card is a longer chain and a higher mode index, not a larger dense grid. The hardness that grows with depth is the many-body content of H (A1: classical death at $(k \geq 8)$), which is exactly the sector QTT of a single Fourier mode never enters.

7. Limits

- Hydrodynamic class: advection–diffusion + interacting medium, not incompressible NS.
- Design-model waveforms used in the local regrade are the exact $nr=10$ closed orbit; the flight is $nr=21$. One damping envelope absorbs size and device damping (A1-P). Box $nr=21$ anchors were cited in the ledger and are not in this folder; we do not pretend they are.
- Frequency grid is coarse (nine irregular depths). Discriminator = sidebands and time-domain revivals, not a high-resolution lineshape.
- $P4 / X(q,k)$ extracted from retrieved counts of this job (see `qlga_xqk.json`). The extract is the fluctuation channel of the interacting medium, not incompressible Navier–Stokes. $(k \geq 8)$ inherits the A1 class; we do not re-adjudicate classical death on this file.
- Single vendor, single flight. No sealed multi-formulation screen, no second backend replica of this card.
- Jennings [9] bound: this is a route advantage, not an unbounded solver.

8. Reproducibility

```
`` python qlga_advantage.py # local grade; writes qlga_advantage.json python verify.py # headline numbers
from archived files python airbus_qlga.py design # statevector gate (no credentials) python airbus_qlga.py
retrieve # IBM; banked job only, no new submit python airbus_qlga.py xqk # refuses unless counts are on
disk python qtt_fv.py g1 # annex cross-check (CPU) ``
```

Every hardware number in this document is from `d9solsgpdb6s73e6a6q0 / d9solbntfhrrs73dt1t40`. The operator constant is the engine's derived operator constant `MU_DERIVED`.

Team Merlin Digital — the route, the bound, and the receipts.